R20

Code No.: 20CSC20

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. V Sem (Main & Backlog) Examination Dec 2023 – Jan 2024

Operating Systems

(Common to CSE, CSM, CET)

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A & Part –B (Internal Choice)** at one place in the same order

Part - A
(5Q X 3M = 15 Marks)

			M	CO	BT
1		Some CPUs provide for more than two modes of operation. What are two possible uses of these multiple modes?	(3)	1	2
2		State whether you agree with the following statement on the basis of what you read in operating system: "Concurrency increases the scheduling overhead without providing any speedup of an application program". Justify.	(3)	2	2
3		The BTV operating system has a 21-bit virtual address, yet on certain embedded devices, it has only a 16-bit physical address. It also has a 2-KB page size. How many entries are there in each of the (i) A conventional, single-level page table and (ii) An inverted page table.	(3)	3	3
4		List the security attacks that cannot be prevented by encryption.	(3)	4	1
5		What are the interesting Signals in Unix process management?	(3)	5	1

Part – B
(5Q X 9M = 45 Marks)

			M	CO	BT
6	(a)	What are the three main purposes of an operating system? What is the main difficulty that a programmer must overcome in writing an operating system for a real-time environment?	(5)	1	2
	(b)	What is the main advantage of the microkernel approach to system design? How do user programs and system services interact in a microkernel architecture? What are the disadvantages of using the microkernel approach?	(4)	1	2
		(OR)			
7	(a)	It is sometimes difficult to achieve a layered approach if two components of the operating system are dependent on each other. Identify a scenario in which it is unclear, how to layer the two system components that require tight coupling of their functionalities.	(4)	1	3
	(b)	Which of the following operating systems has the highest portability? Provide the proper reasons for your answer. i) An OS with a monolithic structure. ii) An OS with a layered structure. iii) A virtual machine OS, iv) A kernel-based OS.	(5)	1	3
8	(a)	A process is in the blocked swapped state. i. Give a sequence of state transitions through which it could have reached this state. ii. Give a sequence of state transitions through which it can reach the ready state. Is more than one sequence of state transitions possible in each of these cases?	(5)	2	3

	(b)	List the criteria for scheduling algorithms. Which CPU scheduling algorithm gives better response time?	(4)	2	3

(OR)

9	(a)	Show that when the banker's algorithm is applied to a finite set of processes, each having a finite execution time, each resource request will be granted eventually.	(5)	2	3
	(b)	Let us consider that there are three processes (P1, P2, P3) in a system and their CPU burst times are shown in the following table.	(4)	2	3

Process	Arrival time	CPU burst time
P1	1	8
P2	0	2
P3	0	1

i) Draw the Gant charts for FCFS, Round Robin algorithm (Quantum time = 2).

ii) Compute the average time and turnaround time for both the algorithms.

10	(a)	Explain the following memory management techniques. i) Paging ii) Segmentation	(5)	3	3
	(b)	List and briefly discuss about various page table organization techniques.	(4)	3	2

(OR)

11	(a)	A file system uses indexed allocation of disk space; however, it permits a sequential file to contain partially full disk blocks. What are the advantages and disadvantages of this scheme?	(4)	3	2
	(b)	Compare the throughput achieved by a RAID level 5 organization with that achieved by a RAID level 1 organization for the following: i. Read operations on single blocks. ii. Read operations on multiple contiguous blocks.	(5)	3	2
12	(a)	A password may become known to other users in a variety of ways. Is there a simple method for detecting that such an event has occurred? Explain your answer.	(4)	4	2
	(b)	Discuss about program threats.	(5)	4	2

(OR)

13	(a)	Write short note on viruses, worms, and rootkits.	(5)	4	2
	(b)	Provide an illustrative description on OS-level memory protection.	(4)	4	2
14	(a)	Provide the illustrative description on the Process state transitions in Unix.	(5)	5	2
	(b)	Discuss about various Linus modules.	(4)	5	2

(OR)

15	(a)	Write about Page Replacement and Swapping mechanisms in Unix.	(5)	5	2
	(b)	Describe about Unix inode structure.	(4)	5	2

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